

Geographic Information Systems

Science

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**SE
TU**

Ollscoil
Teicneolaíochta
an Oirdheiscirt

South East
Technological
University

Geographic Information Systems (A11812)

Short Title: Geographic Information Systems
Faculty Science and Computing
Department Science
Cluster Forestry
Author MPEDINI
Credits 5

Level: Advanced

Description of Module / Aims

The very nature of forestry dictates that forest information systems are mainly concerned with the storage, maintenance, analysis and presentation of area-related data. Geographic Information Systems (GIS) along with database software form the corner stones of a forest information system. In this module students will build on their skills in cartography developed in the 2nd year module Applied GIS. The module uses the commercial software package ArcGIS for Desktop. Students will plan their GIS projects, perform basic spatial analysis on vector/raster data sets, and work with spatial estimation and modelling. Additionally a 3D visualization will be carried out for a specific forestry topic.

Programmes

INF0-0009 BSc (Hons) in Land Management (in Forestry) (WD_SBSFO_B)

1 / 2 / M

Indicative Content

- Raster and vector data models
- Geodesy, datums, projected and geographic coordinate systems
- Attribute tables
- Geodatabase design
- Querying data
- Fundamental spatial analysis
- Basic raster analysis
- Terrain analysis
- Spatial estimation: Interpolation, prediction and core area delineation
- Spatial models and modelling
- Data quality

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Discriminate between the main characteristics and suitability of vector and raster data models.
2. Solve map projection issues for vector and raster data in advance of starting a GIS project.
3. Formulate the requirements to spatial data sets and design a geodatabase in order to solve specific GIS problems.
4. Modify database content in ArcGIS attribute tables by building query expressions.
5. Solve basic spatial problems using analysis such as selection, buffering, dissolve, overlay and classification on both vector and raster datasets.
6. Develop spatial models in the ArcGIS Modelbuilder incorporating terrain analysis and spatial estimation.

Learning and Teaching Methods

- Lectures will be used to present the concepts of spatial data and spatial analysis.
- In practicals students will solve realistic spatial problems using ArcGIS for Desktop, both guided and independently.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
Assignment	25%	1,2,3,4
Assignment	25%	4,5,6
Assignment	50%	1,2,3,4,5,6

Assessment Criteria

- <40%:** Very limited knowledge and understanding of the subject matter. Lacking in technical ability/skill and understanding to carry out practical/professional tasks. Failure to meet the objectives of projects. Unable to carry out or submit independent study and research.
- 40%-49%:** Demonstrate a limited knowledge of the subject matter. Show some technical ability/skill in carrying out and reporting on practical/professional tasks. Attempt made to meet objectives and carry out independent study and research. Final result lacks balance and arguments undeveloped.
- 50%-59%:** Demonstrate satisfactory general knowledge of the main issues within the subject. Logically and competently carry out practical/professional tasks. Satisfactory handling and awareness in carrying out independent study and research. Reporting and projects adequately structured and referenced.
- 60%-69%:** Demonstrate sound knowledge of the subject matter. Show ability to analyse and logically argue in an effective and mature style. Demonstrate initiative and the ability to criticise methods used and appreciation of experimental design when carrying out practical/professional tasks. Projects and reporting coherent and soundly structured illustrating wide and in depth reading on subject coupled with the proper use of references.
- 70%-100%:** Demonstrate authoritative handling of complex material and a highly developed and extensive understanding of the subject. Provide well focused analysis and convincing argument on the subject. Demonstrate imaginative approach, excellent technical ability, awareness, and aptitude in carrying out and reporting on practical/professional tasks. Show initiative, individual thought, and enthusiasm in self study and independent learning.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	12	
Practical	36	
Independent Learning	87	

Essential Material

- "ESRI." www.esri.com
- "Ordnance Survey Ireland." www.osi.ie

- Bolstad, P. _GIS Fundamentals. A First Text on Geographic Information Systems_. 4th ed. MN US: Eider Press, 2012.

Supplementary Material

- Madej, E. _Cartographic Design using ArcView GIS_. NY, USA: OnWord Press, 2001.

Requested Resources

- Room Type: Computer Lab